

12 Vector-Valued Functions



12.2

Differentiation and Integration of Vector-Valued Functions

Objectives

- Differentiate a vector-valued function.
- Integrate a vector-valued function.



Differentiation of Vector-Valued Functions

Differentiation of Vector-Valued Functions

The definition of the derivative of a vector-valued function parallels the definition for real-valued functions.

Definition of the Derivative of a Vector-Valued Function

The derivative of a vector-valued function $\mathbf{r}$ is

$$\mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

for all t for which the limit exists. If $\mathbf{r}'(t)$ exists, then $\mathbf{r}$ is **differentiable at t** . If $\mathbf{r}'(t)$ exists for all t in an open interval I , then $\mathbf{r}$ is **differentiable on the interval I** . Differentiability of vector-valued functions can be extended to closed intervals by considering one-sided limits.

Differentiation of Vector-Valued Functions

Differentiation of vector-valued functions can be done on a *component-by-component basis*. To see why this is true, consider the function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$.

Applying the definition of the derivative produces the following.

$$\begin{aligned}\mathbf{r}'(t) &= \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \\&= \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t)\mathbf{i} + g(t + \Delta t)\mathbf{j} - f(t)\mathbf{i} - g(t)\mathbf{j}}{\Delta t} \\&= \lim_{\Delta t \rightarrow 0} \left\{ \left[\frac{f(t + \Delta t) - f(t)}{\Delta t} \right] \mathbf{i} + \left[\frac{g(t + \Delta t) - g(t)}{\Delta t} \right] \mathbf{j} \right\} \\&= \left\{ \lim_{\Delta t \rightarrow 0} \left[\frac{f(t + \Delta t) - f(t)}{\Delta t} \right] \right\} \mathbf{i} + \left\{ \lim_{\Delta t \rightarrow 0} \left[\frac{g(t + \Delta t) - g(t)}{\Delta t} \right] \right\} \mathbf{j} \\&= f'(t)\mathbf{i} + g'(t)\mathbf{j}\end{aligned}$$

Differentiation of Vector-Valued Functions

This important result is listed in the theorem shown below.

THEOREM 12.1 Differentiation of Vector-Valued Functions

1. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, where f and g are differentiable functions of t , then

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j}. \quad \text{Plane}$$

2. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f , g , and h are differentiable functions of t , then

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}. \quad \text{Space}$$

Differentiation of Vector-Valued Functions

Note that the derivative of the vector-valued function $\mathbf{r}$ is itself a vector-valued function.

You can see from Figure 12.8 that $\mathbf{r}'(t)$ is a vector tangent to the curve given by $\mathbf{r}(t)$ and pointing in the direction of increasing t -values.

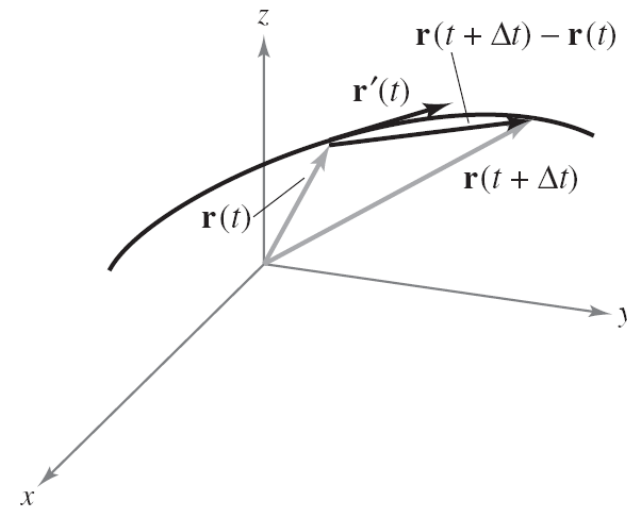


Figure 12.8

Example 1 – *Differentiation of a Vector-Valued Function*

For the vector-valued function $\mathbf{r}(t) = t\mathbf{i} + (t^2 + 2)\mathbf{j}$ find $\mathbf{r}'(t)$. Then sketch the plane curve represented by $\mathbf{r}(t)$ and the graphs of $\mathbf{r}(1)$ and $\mathbf{r}'(1)$.

Solution:

Differentiate on a component-by-component basis to obtain

$$\mathbf{r}'(t) = \mathbf{i} + 2t \mathbf{j}. \quad \text{Derivative}$$

From the position vector $\mathbf{r}(t)$, you can write the parametric equations $x = t$ and $y = t^2 + 2$.

The corresponding rectangular equation is $y = x^2 + 2$.

When $t = 1$, $\mathbf{r}(1) = \mathbf{i} + 3\mathbf{j}$ and $\mathbf{r}'(1) = \mathbf{i} + 2\mathbf{j}$.

Example 1 – *Solution*

cont'd

In Figure 12.9, $\mathbf{r}(1)$ is drawn starting at the origin, and $\mathbf{r}'(1)$ is drawn starting at the terminal point of $\mathbf{r}(1)$. Note that at $(1, 3)$, the vector $\mathbf{r}'(1)$ is tangent to the curve given by $\mathbf{r}(t)$ and is pointing in the direction of increasing t -values.

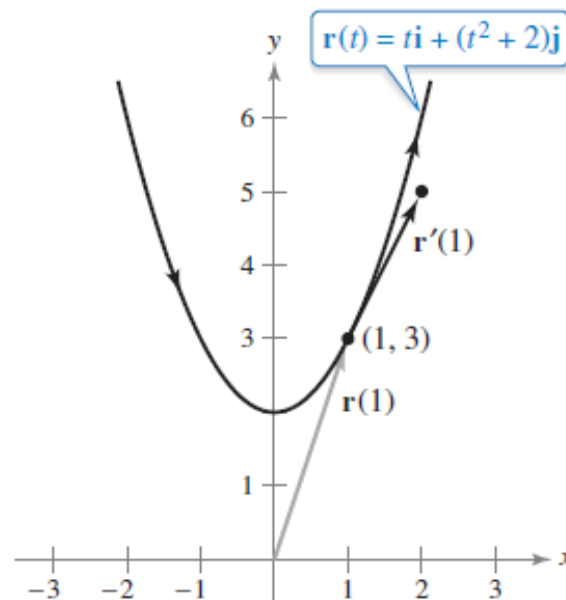


Figure 12.9
Dr P. Patel

Differentiation of Vector-Valued Functions

The parametrization of the curve represented by the vector-valued function

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

is **smooth on an open interval I** when f' , g' , and h' are continuous on I and $\mathbf{r}'(t) \neq \mathbf{0}$ for any value of t in the interval I .

Example 3 – *Finding Intervals on Which a Curve Is Smooth*

Find the intervals on which the epicycloid C given by

$$\mathbf{r}(t) = (5 \cos t - \cos 5t)\mathbf{i} + (5 \sin t - \sin 5t)\mathbf{j}, \quad 0 \leq t \leq 2\pi$$

is smooth.

Solution:

The derivative of $\mathbf{r}$ is

$$\mathbf{r}'(t) = (-5 \sin t + 5 \sin 5t)\mathbf{i} + (5 \cos t - 5 \cos 5t)\mathbf{j}.$$

In the interval $[0, 2\pi]$, the only values of t for which

$$\mathbf{r}'(t) = 0\mathbf{i} + 0\mathbf{j}$$

are $t = 0, \pi/2, \pi, 3\pi/2$, and 2π .

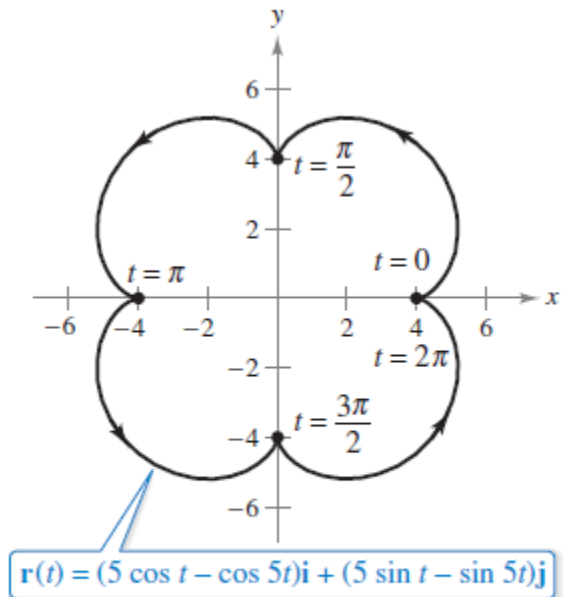
Example 3 – *Solution*

cont'd

Therefore, you can conclude that C is smooth on the intervals

$$\left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right), \left(\pi, \frac{3\pi}{2}\right), \text{ and } \left(\frac{3\pi}{2}, 2\pi\right)$$

as shown in Figure 12.10.



The epicycloid is not smooth at the points where it intersects the axes.

Figure 12.10

Differentiation of Vector-Valued Functions

In Figure 12.10, note that the curve is not smooth at points at which the curve makes abrupt changes in direction.

Such points are called *cusps* or **nodes**.

Differentiation of Vector-Valued Functions

THEOREM 12.2 Properties of the Derivative

Let $\mathbf{r}$ and $\mathbf{u}$ be differentiable vector-valued functions of t , let w be a differentiable real-valued function of t , and let c be a scalar.

1. $\frac{d}{dt}[c\mathbf{r}(t)] = c\mathbf{r}'(t)$
2. $\frac{d}{dt}[\mathbf{r}(t) \pm \mathbf{u}(t)] = \mathbf{r}'(t) \pm \mathbf{u}'(t)$
3. $\frac{d}{dt}[w(t)\mathbf{r}(t)] = w(t)\mathbf{r}'(t) + w'(t)\mathbf{r}(t)$
4. $\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{u}(t)] = \mathbf{r}(t) \cdot \mathbf{u}'(t) + \mathbf{r}'(t) \cdot \mathbf{u}(t)$
5. $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{u}(t)] = \mathbf{r}(t) \times \mathbf{u}'(t) + \mathbf{r}'(t) \times \mathbf{u}(t)$
6. $\frac{d}{dt}[\mathbf{r}(w(t))] = \mathbf{r}'(w(t))w'(t)$
7. If $\mathbf{r}(t) \cdot \mathbf{r}(t) = c$, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$.

Example 4 – *Using Properties of the Derivative*

For $\mathbf{r}(t) = \frac{1}{t}\mathbf{i} - \mathbf{j} + \ln t\mathbf{k}$ and $\mathbf{u}(t) = t^2\mathbf{i} - 2t\mathbf{j} + \mathbf{k}$, find each derivative.

a. $\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{u}(t)]$

b. $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{u}'(t)].$

Example 4(a) – *Solution*

Because $\mathbf{r}'(t) = -\frac{1}{t^2}\mathbf{i} + \frac{1}{t}\mathbf{k}$ and $\mathbf{u}'(t) = 2t\mathbf{i} - 2\mathbf{j}$, you have

$$\begin{aligned}\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{u}(t)] &= \mathbf{r}(t) \cdot \mathbf{u}'(t) + \mathbf{r}'(t) \cdot \mathbf{u}(t) \\&= \left(\frac{1}{t}\mathbf{i} - \mathbf{j} + \ln t\mathbf{k}\right) \cdot (2t\mathbf{i} - 2\mathbf{j}) + \left(-\frac{1}{t^2}\mathbf{i} + \frac{1}{t}\mathbf{k}\right) \cdot (t^2\mathbf{i} - 2t\mathbf{j} + \mathbf{k}) \\&= 2 + 2 + (-1) + \frac{1}{t} \\&= 3 + \frac{1}{t}.\end{aligned}$$

Example 4(b) – *Solution*

cont'd

Because $\mathbf{u}'(t) = 2t\mathbf{i} - 2\mathbf{j}$ and $\mathbf{u}''(t) = 2\mathbf{i}$, you have

$$\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{u}'(t)] = [\mathbf{u}(t) \times \mathbf{u}''(t)] + [\mathbf{u}'(t) \times \mathbf{u}'(t)]$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ t^2 & -2t & 1 \\ 2 & 0 & 0 \end{vmatrix} + \mathbf{0}$$

$$= \begin{vmatrix} -2t & 1 \\ 0 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} t^2 & 1 \\ 2 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} t^2 & -2t \\ 2 & 0 \end{vmatrix} \mathbf{k}$$

$$= 0\mathbf{i} - (-2)\mathbf{j} + 4t\mathbf{k}$$

$$= 2\mathbf{j} + 4t\mathbf{k}.$$



Integration of Vector-Valued Functions

Integration of Vector-Valued Functions

The following definition is a consequence of the definition of the derivative of a vector-valued function.

Definition of Integration of Vector-Valued Functions

1. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, where f and g are continuous on $[a, b]$, then the **indefinite integral (antiderivative)** of $\mathbf{r}$ is

$$\int \mathbf{r}(t) dt = \left[\int f(t) dt \right] \mathbf{i} + \left[\int g(t) dt \right] \mathbf{j} \quad \text{Plane}$$

and its **definite integral** over the interval $a \leq t \leq b$ is

$$\int_a^b \mathbf{r}(t) dt = \left[\int_a^b f(t) dt \right] \mathbf{i} + \left[\int_a^b g(t) dt \right] \mathbf{j}.$$

2. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f , g , and h are continuous on $[a, b]$, then the **indefinite integral (antiderivative)** of $\mathbf{r}$ is

$$\int \mathbf{r}(t) dt = \left[\int f(t) dt \right] \mathbf{i} + \left[\int g(t) dt \right] \mathbf{j} + \left[\int h(t) dt \right] \mathbf{k} \quad \text{Space}$$

and its **definite integral** over the interval $a \leq t \leq b$ is

$$\int_a^b \mathbf{r}(t) dt = \left[\int_a^b f(t) dt \right] \mathbf{i} + \left[\int_a^b g(t) dt \right] \mathbf{j} + \left[\int_a^b h(t) dt \right] \mathbf{k}.$$

Integration of Vector-Valued Functions

The antiderivative of a vector-valued function is a family of vector-valued functions all differing by a constant vector $\mathbf{C}$.

For instance, if $\mathbf{r}(t)$ is a three-dimensional vector-valued function, then for the indefinite integral $\int \mathbf{r}(t) dt$, you obtain three constants of integration

$$\int f(t) dt = F(t) + C_1, \quad \int g(t) dt = G(t) + C_2, \quad \int h(t) dt = H(t) + C_3$$

where $F'(t) = f(t)$, $G'(t) = g(t)$, and $H'(t) = h(t)$.

Integration of Vector-Valued Functions

These three *scalar* constants produce one *vector* constant of integration

$$\begin{aligned}\int \mathbf{r}(t) \, dt &= [F(t) + C_1]\mathbf{i} + [G(t) + C_2]\mathbf{j} + [H(t) + C_3]\mathbf{k} \\ &= [F(t)\mathbf{i} + G(t)\mathbf{j} + H(t)\mathbf{k}] + [C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}] \\ &= \mathbf{R}(t) + \mathbf{C}\end{aligned}$$

where $\mathbf{R}'(t) = \mathbf{r}(t)$.

Example 5 – *Integrating a Vector-Valued Function*

Find the indefinite integral

$$\int (t\mathbf{i} + 3\mathbf{j}) \, dt.$$

Solution:

Integrating on a component-by-component basis produces

$$\int (t\mathbf{i} + 3\mathbf{j}) \, dt = \frac{t^2}{2}\mathbf{i} + 3t\mathbf{j} + \mathbf{C}.$$